

HANOI UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY



Group: 6

Assignment title: Explain Video Engagement Patterns on the MixiGaming Channel Using Interpretable Regression Models

Class: 01.2023C

Course title: BDM Course

Submission date: 06/05/2026

Members:

Đào Hải Nam: 2301140064

Nguyễn Đăng Hải: 2301140028

Nguyễn Thanh Minh: 2301140063

Table of Content

Abstract.....	3
I. Introduction.....	3
II. Related Work.....	4
III. Methodology.....	5
3.1 Dataset Description.....	5
3.2 Data Preprocessing.....	7
3.3 Exploratory Data Analysis and Hypothesis Formulation.....	8
3.4 Regression Models.....	14
3.5 Evaluation Strategy.....	15
IV. Implementation and Results.....	16
4.1 Experimental Setup.....	16
4.2 Dataset Statistics and Preliminary Evidence.....	17
4.3 Performance Comparison.....	17
4.4 Overfitting and Generalization Analysis.....	18
4.5 Explainability Results.....	20
4.6 Decision Rules and Hypothesis Verification.....	23
4.7 Sample Evidence from Real Data.....	25
V. Discussion.....	26
VI. Conclusion.....	27
References.....	27

EXPLAINING VIDEO ENGAGEMENT PATTERNS ON THE MIXIGAMING CHANNEL USING INTERPRETABLE REGRESSION MODELS

Abstract

Viewer engagement on YouTube is an important topic in data mining because metrics such as views, likes, and comments reflect audience attention, popularity, and interaction. This study investigates engagement patterns on the MixiGaming YouTube channel by examining how selected video-related and temporal features are associated with three engagement outcomes: views, likes, and comments. The dataset consists of 730 publicly available videos collected from the 2024–2025 period using the YouTube Data API v3. The analysis applies exploratory data analysis, feature engineering, log-transformed regression modeling, baseline comparison, and SHAP-based model explainability.

The results show that engagement metrics are strongly right-skewed, supporting the use of log transformation before model training. Although the Decision Tree Regressor produced limited test-set generalization and negative R^2 values, the comparison with Linear Regression provides useful evidence that the selected interpretable feature set has weak predictive power under temporal testing conditions. However, feature importance and SHAP analysis reveal consistent explanatory patterns: previous video performance is the strongest predictor across views, likes, and comments, followed by video duration and monthly temporal variation. Overall, the study suggests that MixiGaming engagement is partly shaped by channel momentum, while upload frequency and upload-hour grouping have weaker explanatory roles. The findings should therefore be interpreted as feature-level associations rather than causal or highly accurate forecasting results.

I. Introduction

The rapid expansion of online video platforms has made viewer engagement analysis an important research topic in data mining and digital media studies. YouTube provides a rich environment for this analysis because it offers publicly observable video-level metadata and interaction indicators. Metrics such as view count, like count, and comment count are commonly used as indicators of audience attention, popularity, and interaction.

Prior studies have shown that YouTube engagement is shaped not only by video content but also by temporal patterns, content-agnostic features, publishing context, and presentation cues [1]–[8]. However, interpretability-focused studies on Vietnamese YouTube channels, especially in the gaming domain, remain limited. Recent work such as EnTube demonstrates that engagement modeling on Vietnamese YouTube data is feasible, but it focuses on multi-channel and multimodal prediction [10]. By contrast, this report focuses on a single Vietnamese gaming channel and prioritizes interpretable regression over large-scale predictive complexity.

The MixiGaming channel is selected as the case study because it is a major Vietnamese gaming content channel and provides a sufficient number of publicly available videos for structured analysis. This study analyzes 730 MixiGaming videos from the 2024–2025 period and examines three engagement targets: views, likes, and comments. Because the study is designed as an explanatory case study, the model results are interpreted as evidence of feature associations rather than as a strong forecasting system.

The main research question is as follows:

Which video-related and temporal factors are associated with viewer engagement on the MixiGaming YouTube channel, and to what extent can historical data from 2024–2025 provide limited temporal insight into later video performance within the same channel?

To address this question, the study develops a pipeline consisting of data collection, preprocessing, exploratory analysis, regression modeling, and SHAP-based interpretation. The final report is organized as follows. Section II reviews related work. Section III describes the methodology. Section IV presents implementation details and empirical results. Section V discusses the findings and limitations. Section VI concludes the study.

This study makes three main contributions. First, it provides a focused Vietnamese gaming-channel case study, which is less represented in existing YouTube engagement literature. Second, it evaluates whether a compact and interpretable feature set can explain engagement variation under a time-based train-test split. Third, it combines model performance evaluation with SHAP-based interpretation to distinguish between predictive usefulness and explanatory value. This distinction is important because a model may have limited forecasting accuracy while still revealing meaningful associations between video-level features and engagement outcomes.

II. Related Work

Early large-scale studies established that YouTube popularity is highly skewed and evolves over time rather than following a uniform pattern. Cha et al. showed that demand on YouTube is uneven and dynamic, while Szabó and Huberman demonstrated that early access patterns can be informative for estimating longer-term popularity. These studies provide the foundational basis for treating video popularity as a temporal and measurable process rather than a static outcome [1], [2].

Subsequent work refined this temporal perspective. Borghol et al. showed that video popularity depends not only on content itself but also on content-agnostic factors such as prior views and video age [3]. Pinto et al. further showed that early view trajectories contain predictive information beyond simple cumulative counts [4], while Figueiredo et al. analyzed how videos follow different popularity trends over time and how referrers and early signals are associated with later outcomes [5]. Ma et al. proposed a lifetime-aware regression framework, reinforcing the relevance of regression-based modeling for YouTube popularity under temporal variation [6].

A third line of research focuses on observable metadata and non-content drivers. Munaro et al. found that video characteristics and textual presentation factors are associated with likes, comments, and dislikes across more than 11,000 influencer videos [7]. Halim et al. further showed that content-unaware features can substantially influence the popularity potential of YouTube videos across multiple regions [8]. These studies justify the present report's focus on interpretable variables such as upload hour, title type, duration, upload frequency, and previous video performance.

For model interpretation, this report uses SHAP, following Lundberg and Lee, who introduced a unified framework for additive feature attribution and model explanation [9]. This is especially relevant because the purpose of the current project is explanation rather than only prediction. In the Vietnamese context, Le et al. introduced EnTube, a dataset of 23,738 videos from 72 Vietnamese YouTube channels and showed that multimodal features and explainable AI can improve engagement modeling [10]. Compared with EnTube, the current study is narrower in scope but more explicitly focused on single-channel interpretability.

In summary, prior studies show that YouTube engagement is shaped by temporal patterns, content-agnostic features, and metadata-related signals. However, most existing studies either focus on large-scale multi-channel prediction or more complex multimodal modeling. The present study differs by focusing on a single Vietnamese gaming channel and by prioritizing interpretable, feature-level explanation over high-complexity prediction.

Although these studies provide important foundations, several gaps remain relevant to the present work. First, many previous studies emphasize large-scale prediction across multiple channels or platforms, while fewer studies examine whether interpretable models can produce useful explanations within a single-channel context. Second, multimodal approaches can improve predictive accuracy, but they often reduce transparency and require richer data such as thumbnails, transcripts, or video content features. Third, studies on Vietnamese YouTube engagement remain relatively limited, especially for gaming channels where audience behavior may be influenced by streamer loyalty, community routines, and repeated viewing habits. Therefore, this study intentionally focuses on a

smaller but interpretable feature set to examine whether channel-level momentum and temporal metadata can explain engagement patterns in a Vietnamese gaming-channel setting.

III. Methodology

3.1 Dataset Description

The dataset was collected from the MixiGaming YouTube channel using the YouTube Data API v3. The study focuses on videos published during the 2024–2025 period and includes 730 videos. For each video, the dataset contains core metadata such as title, publish date, and video duration, together with engagement statistics including view count, like count, and comment count. Because the data are public and aggregated at the video level, the study does not involve user-level personal information.

To support interpretability-oriented analysis, a limited set of derived variables was constructed. These include title type, upload date, upload hour, video duration in seconds, upload frequency, previous video performance, and month. The restricted feature design reflects the study’s goal of balancing explanatory value with manageable model complexity. The feature design in this study is informed by prior work showing that YouTube popularity can be meaningfully explained by temporal conditions, content-agnostic signals, and metadata-based variables rather than by semantic content alone [3], [6], [7], [8]. Accordingly, the present report restricts the feature space to a theoretically grounded and interpretable subset.

Feature	Description	Type	Role
View count	Number of video views	Numeric	Target
Like count	Number of likes	Numeric	Target
Comment count	Number of comments	Numeric	Target
Duration seconds	Video duration in seconds	Numeric	Predictor

Previous view count	View count of the previous video	Numeric	Predictor
Days since previous upload	Days between consecutive uploads	Numeric	Predictor
Upload hour group	Grouped upload time category	Categorical	Predictor
Title type	Engineered title category	Categorical	Predictor
Month sin / Month cos	Cyclical encoding of month	Numeric	Predictor

Table 1. Summary of the Collected MixiGaming Dataset



Figure 1. Data Collection and Processing Workflow

3.2 Data Preprocessing

The raw dataset underwent preprocessing before modeling. Categorical variables such as title type and upload hour were transformed using one-hot encoding. The month variable was represented through cyclical encoding with sine and cosine transformations in order to capture temporal continuity more appropriately than a purely ordinal treatment. Continuous features, including duration, upload frequency, and previous video views, were normalized using MinMaxScaler.

In addition, two sequential variables were constructed to capture channel dynamics. Upload frequency was derived from the time interval between consecutive uploads, while previous video performance was defined as the view count of the immediately preceding video. These variables were included because engagement on a given video may depend partly on the channel's recent activity and prior audience response. This modeling choice is also compatible with prior temporal-popularity literature showing that early-access information and path-dependent signals can matter for later performance [4]–[6].

The train-test split was implemented using a time-based strategy rather than a random split. Earlier videos were allocated to the training set, while later videos were reserved for the test

set. This design prevents temporal leakage and is more suitable for sequential channel data. The final processed dataset contained 584 training videos and 146 test videos.

Before feature transformation, the dataset was checked for duplicate video records, missing engagement values, and invalid duration values. No videos were removed after cleaning, and the final dataset remained at 730 observations. Since engagement variables such as views, likes, and comments are naturally heavy-tailed, extreme values were not removed as outliers. Instead, the target variables were transformed using $\log(x + 1)$ to reduce skewness while preserving high-performing videos as meaningful observations.

To reduce data leakage, all sequential variables were constructed using only information available before or at the publication order of the current video. In particular, previous video performance was calculated from the immediately preceding video and was not derived from future observations. The time-based train-test split further prevents future videos from influencing model training.

Item	Value
Initial number of videos	730
Videos after cleaning	730
Number of engineered features	7
Training size	584
Test size	146
Encoding method	One-hot / Cyclical / Scaling
Split strategy	Time-based split

Table 2. Preprocessing Outputs and Final Feature Space

3.3 Exploratory Data Analysis and Hypothesis Formulation

Exploratory data analysis was conducted before model training in order to understand the distributional properties of the engagement variables and to identify interpretable patterns worth testing. The three target variables—views, likes, and comments—were examined through descriptive statistics and visual inspection. Initial observation suggested that the

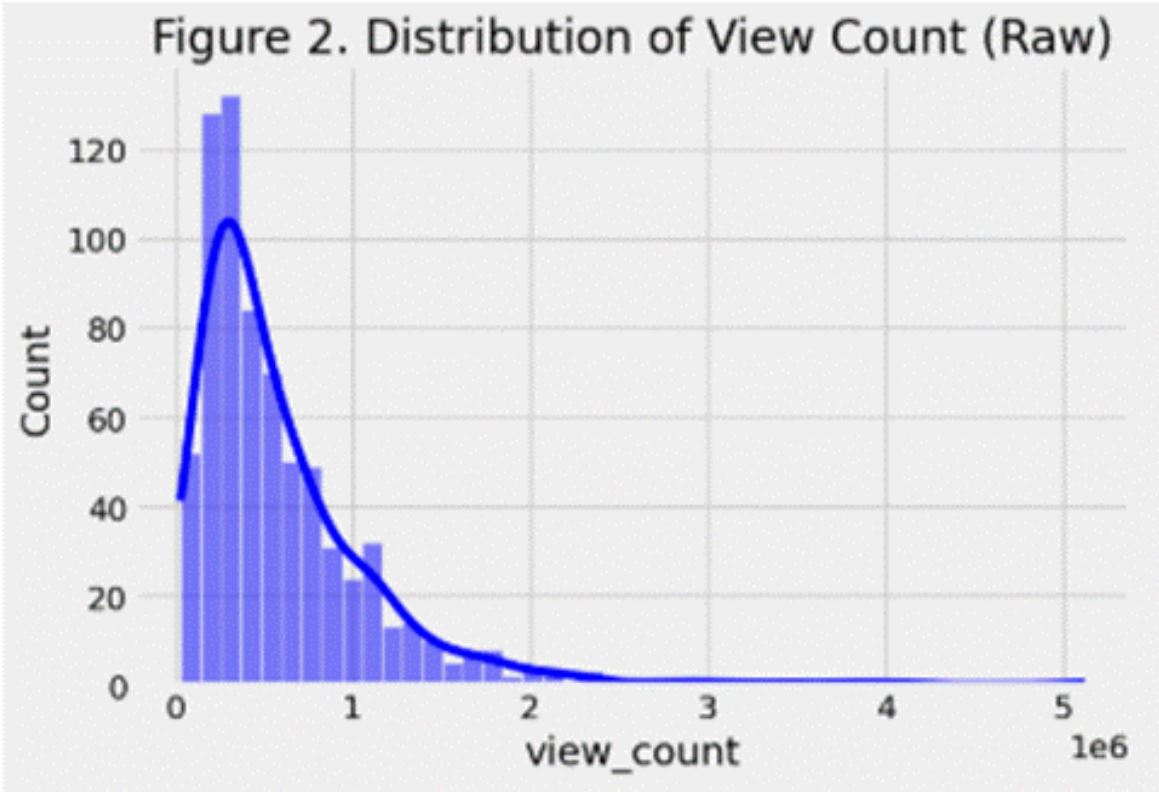
engagement distributions were strongly right-skewed and heavy-tailed, a common property of social media performance variables. This motivated the use of logarithmic transformation for the three target variables before regression modeling.

The exploratory phase also examined group-level differences by title type, upload hour, and temporal month. In addition, the relationship between duration and engagement, and between previous video performance and current outcomes, was inspected. Based on these exploratory findings, the study formulated four core hypotheses: first, videos uploaded in the evening may achieve higher engagement; second, longer videos may be associated with stronger engagement; third, previous video performance may positively relate to current video views; and fourth, upload frequency may have a meaningful relationship with engagement. This emphasis on metadata-related and temporal patterns is consistent with prior work on YouTube popularity drivers [3]–[8].

	View count	Like count	Comment count	Duration second	Previous view count	Days since previous upload
count	7.300000e+02	730.000000	730.000000	730.000000	7.300000e+02	730.000000
mean	6.438932e+05	31887.238356	2658.763014	3807.595890	6.436578e+05	0.998630
std	5.846505e+05	28953.828496	2414.653047	1935.607099	5.846886e+05	0.037012
min	4.274100e+04	2116.000000	176.000000	604.000000	4.274100e+04	0.000000
25%	2.683122e+05	13287.000000	1107.500000	2114.750000	2.683122e+05	1.000000
50%	4.635440e+05	22956.000000	1914.000000	3815.500000	4.633015e+05	1.000000
75%	7.879732e+05	39022.250000	3253.750000	5525.750000	7.879732e+05	1.000000

max	5.691509e+06	281862.00000 0	23506.00 0000	7173.000 000	5.691509 e+06	1.000000
-----	--------------	-------------------	------------------	-----------------	------------------	----------

Table 3. Descriptive Statistics of Engagement Variables



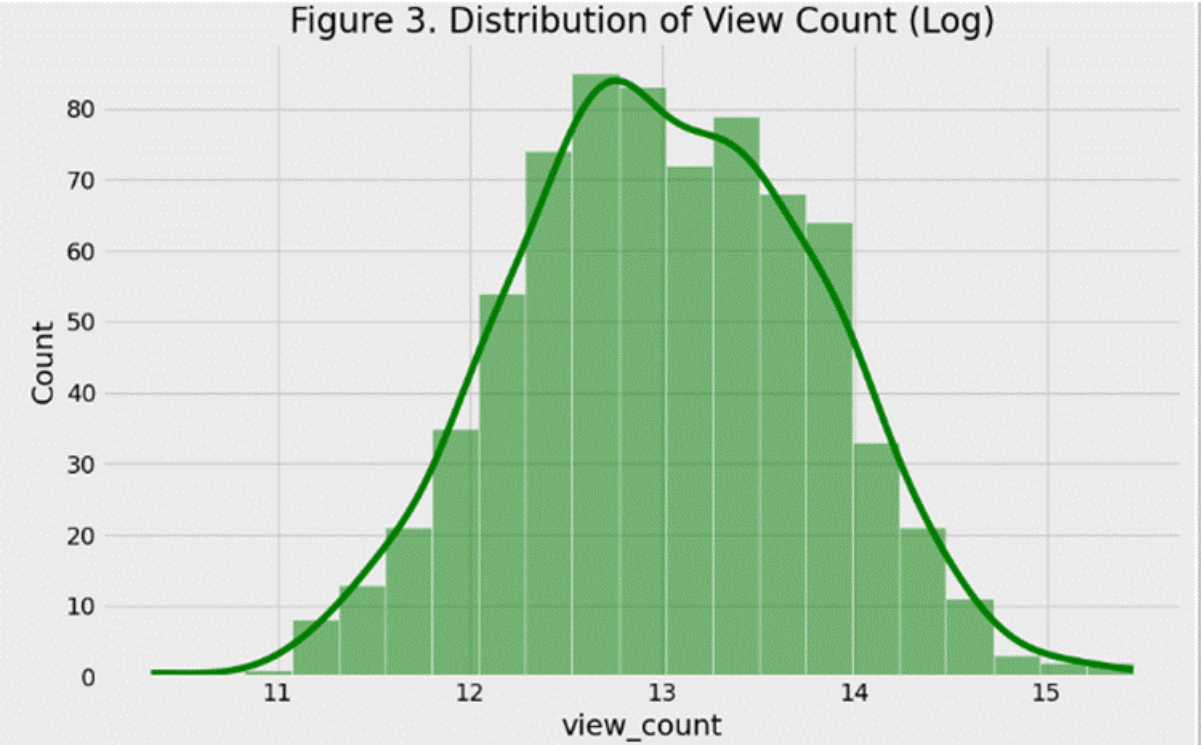
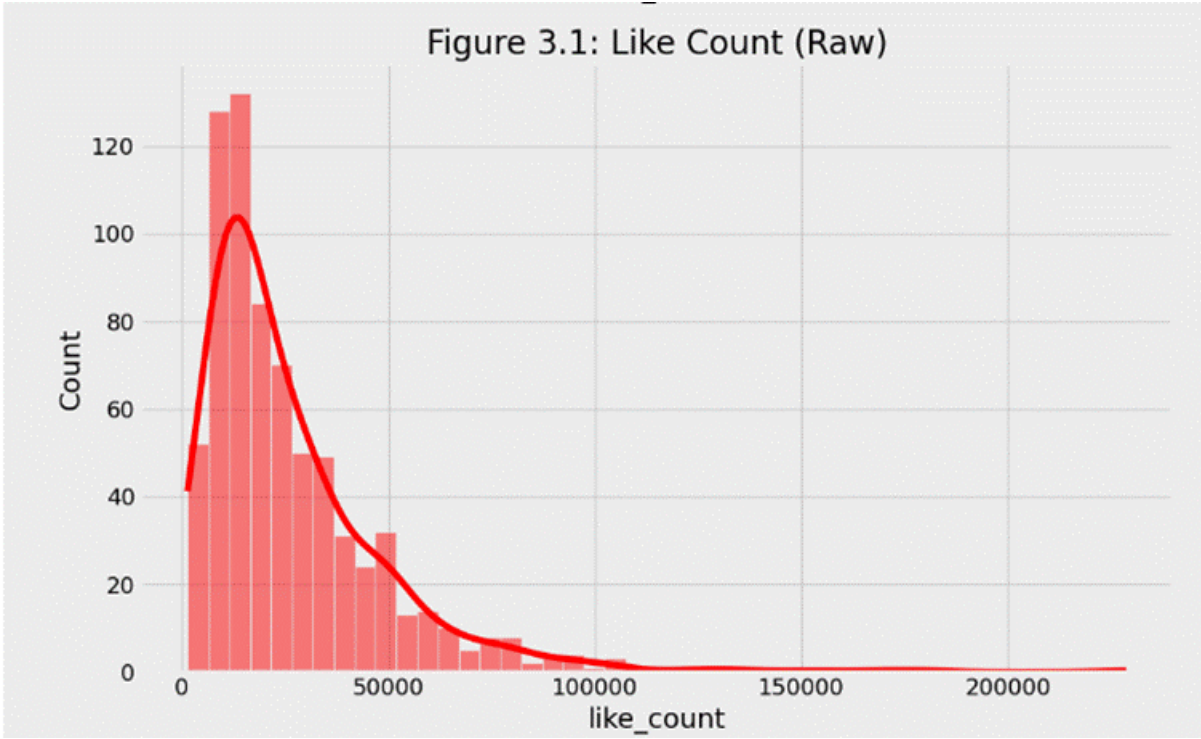


Figure 2. Distribution of View Count



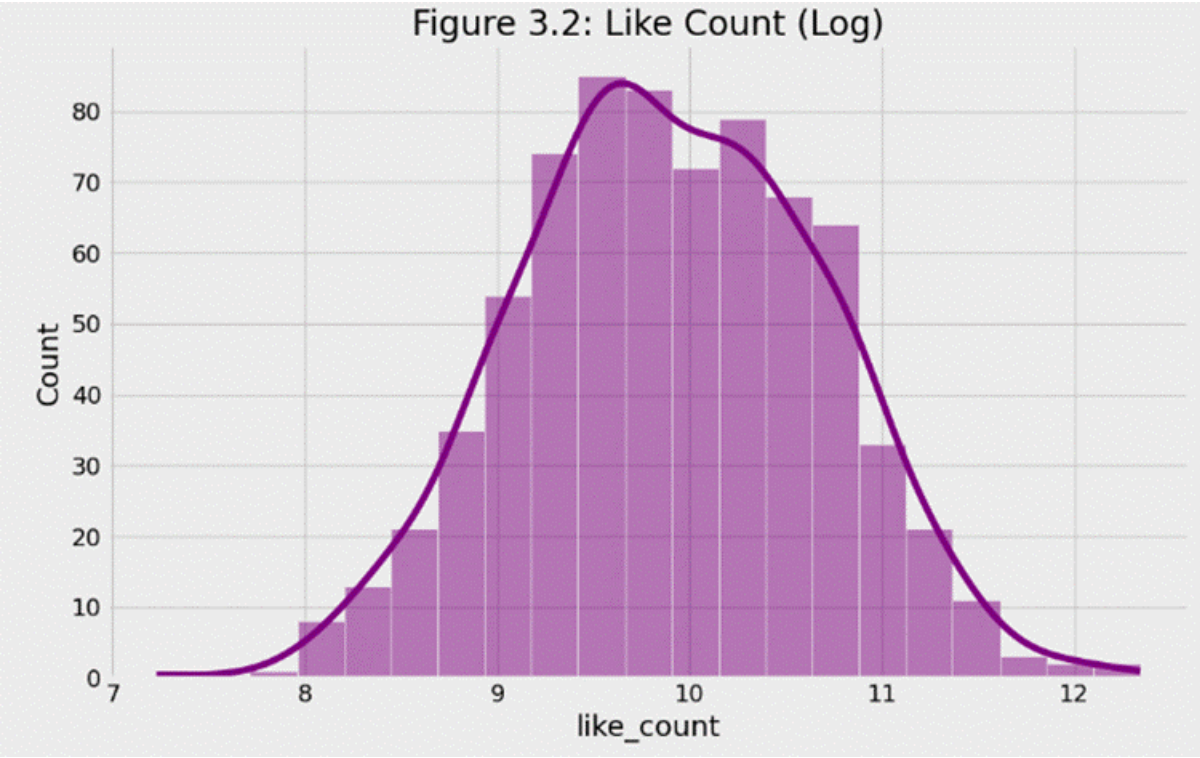
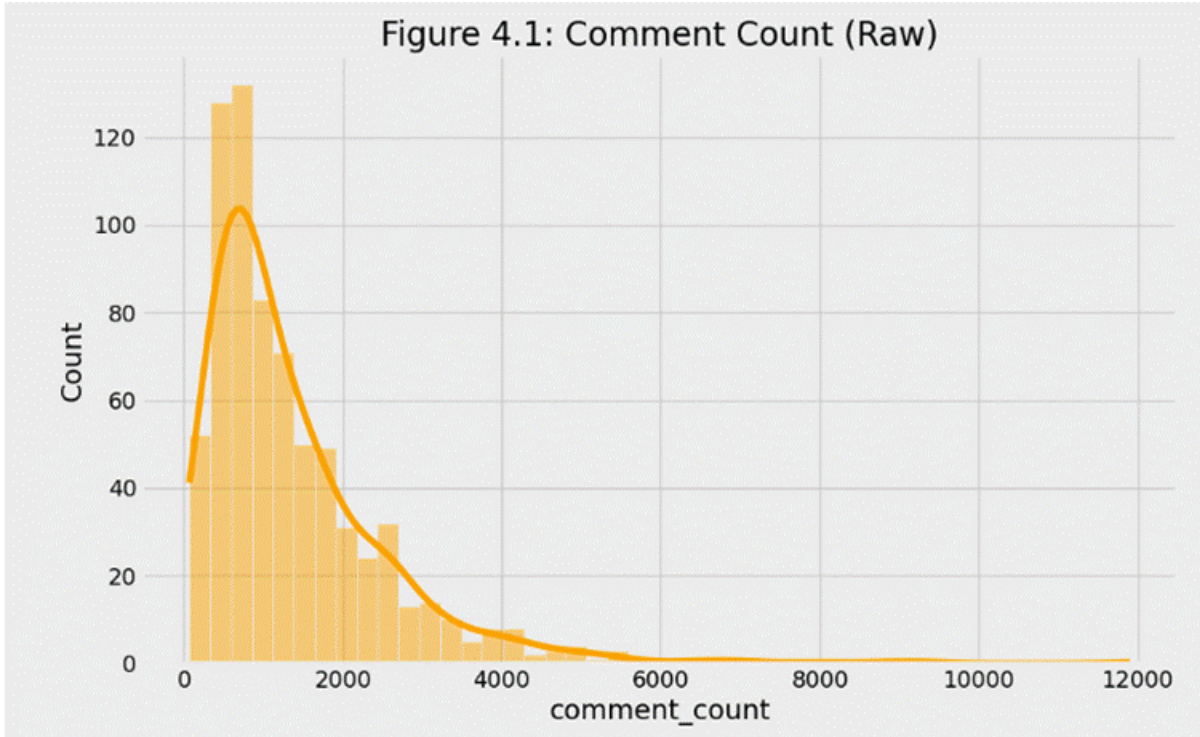


Figure 3. Distribution of Like Count



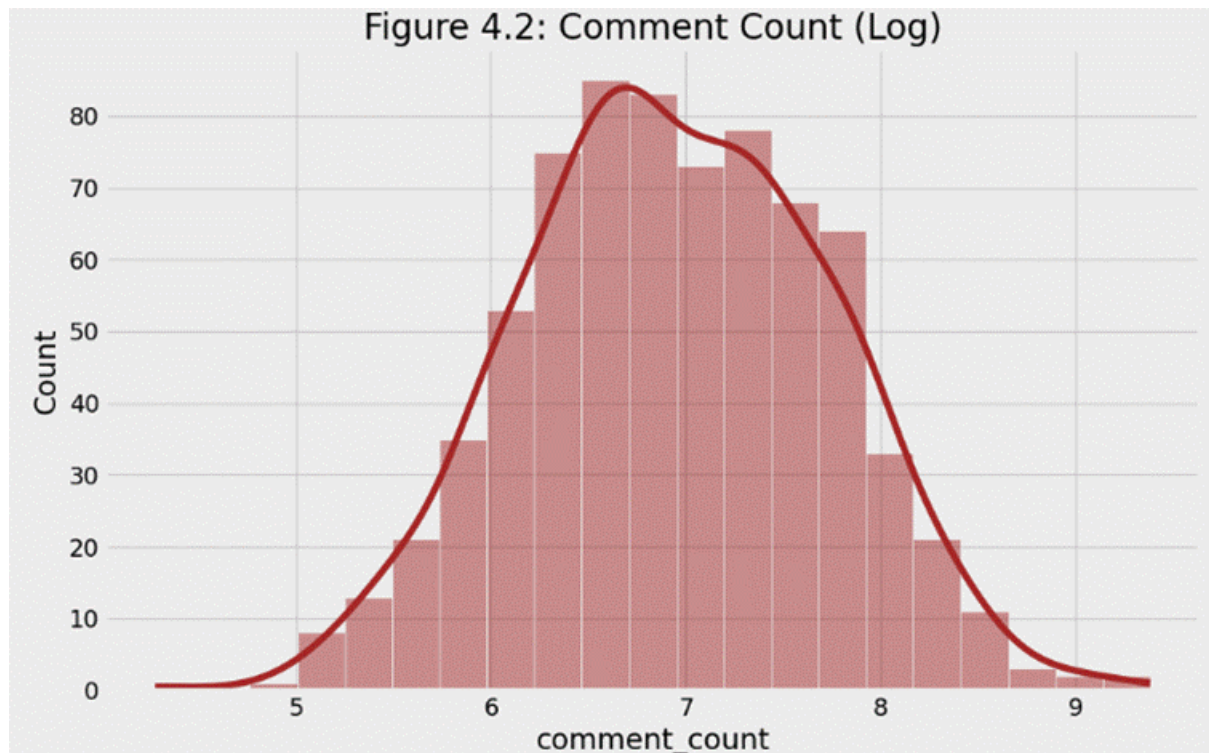


Figure 4. Distribution of Comment Count

Figures 2, 3, and 4 further confirm the long-tailed nature of engagement. The raw distributions of views, likes, and comments are heavily skewed to the right, while the log-transformed distributions are more symmetric and closer to a normal-like shape. This supports the methodological decision to apply $\log(x + 1)$ transformation to the engagement targets before regression modeling.

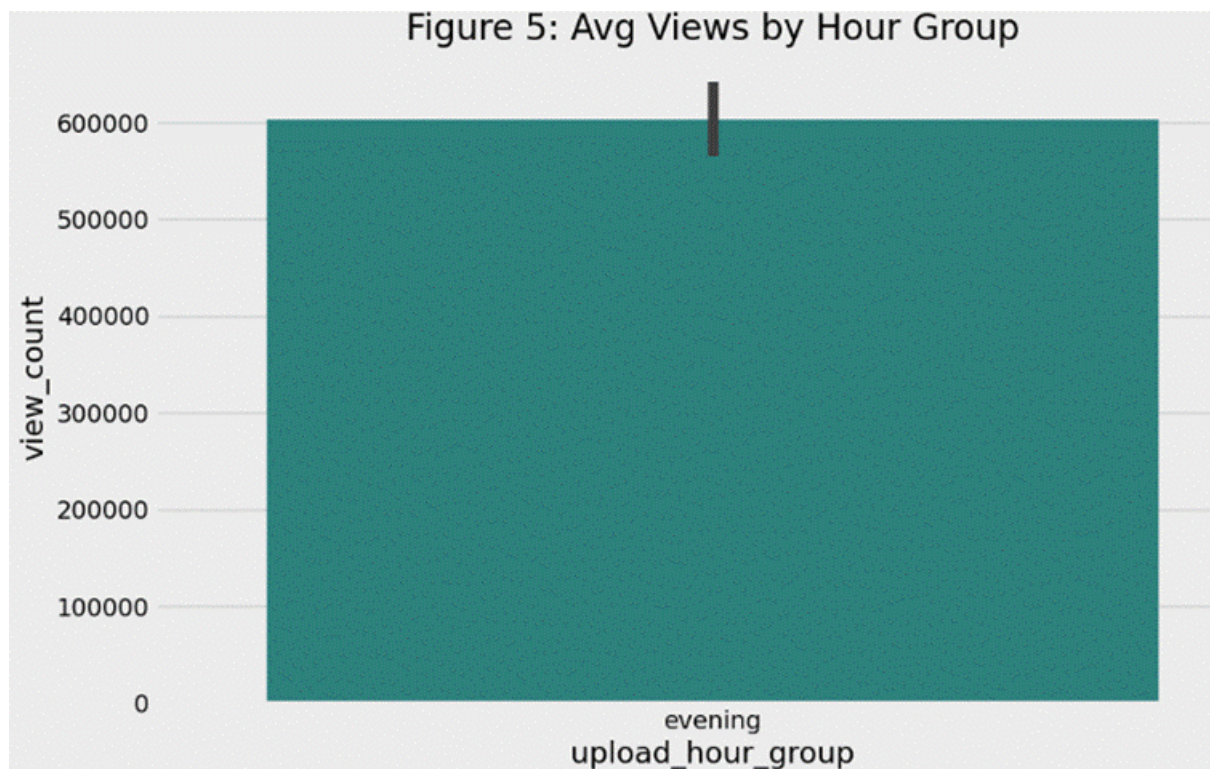


Figure 5. Average Engagement by Upload Hour

Figure 5 shows the average view count by upload-hour group. In the current dataset, uploads are concentrated in the evening group. Therefore, the upload-hour hypothesis should be interpreted cautiously because the available data do not provide a balanced comparison across multiple time groups. The result can still suggest that evening uploads are common for this channel, but it cannot strongly prove that evening uploads outperform other time groups.

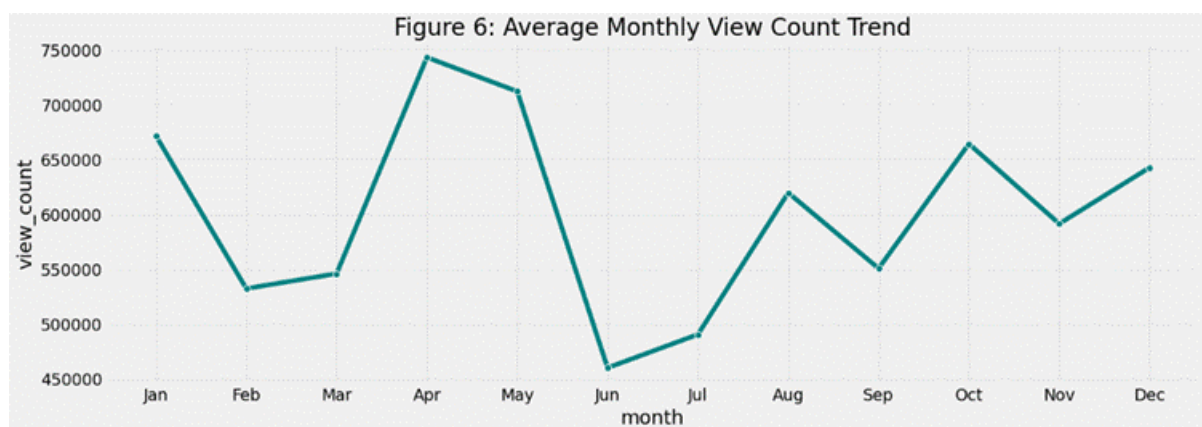


Figure 6. Monthly Engagement Trend

Figure 6 shows the monthly trend of average view count. Engagement varies across months, with visible peaks around April–May and October, and lower performance around June–July. This suggests that month-based temporal variation may be relevant, although it should not be interpreted as a causal seasonal effect without additional contextual variables such as game popularity, content type, or external events.

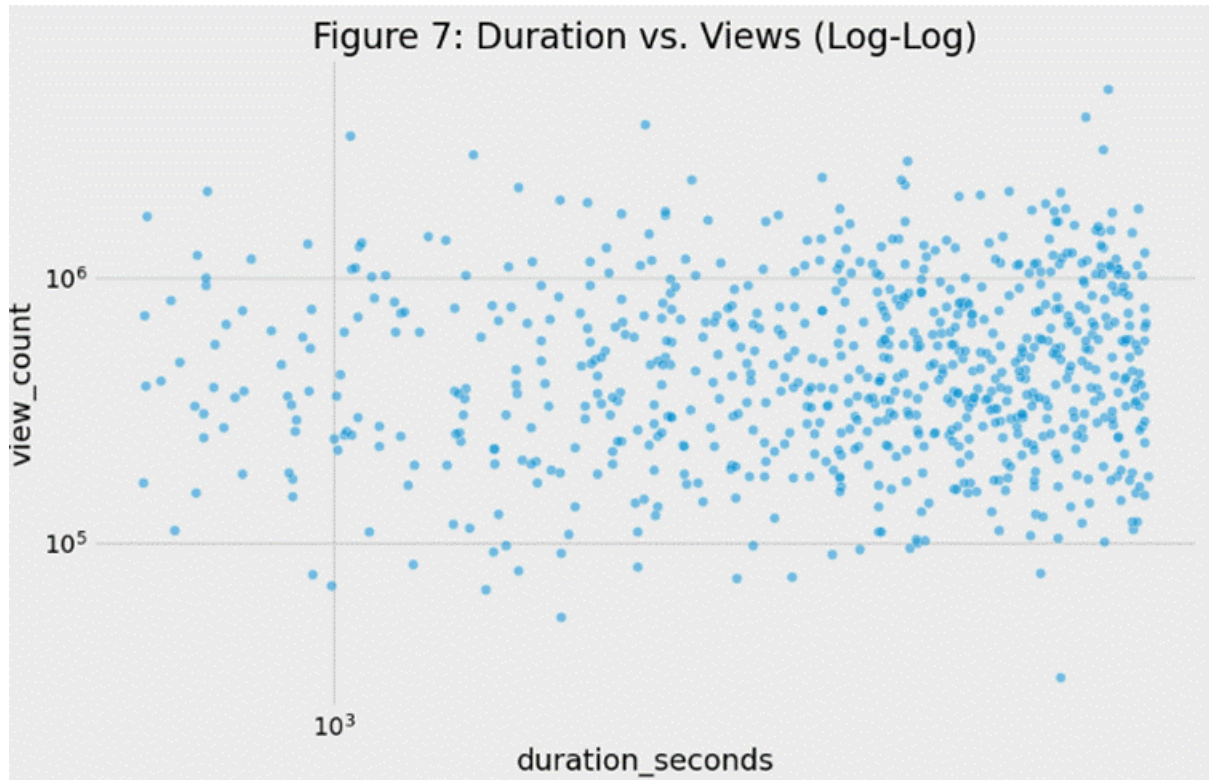


Figure 7. Relationship Between Video Duration and Views

Figure 7 presents the relationship between video duration and views on a log-log scale. The scatter plot does not show a simple linear relationship. Instead, videos with similar durations can receive very different view counts, suggesting that duration alone is insufficient to explain engagement. However, duration may still interact with other factors and contribute to model decisions in a non-linear way.

3.4 Regression Models

The primary model used in this study is the Decision Tree Regressor. This model was selected because it can capture non-linear relationships, detect threshold effects, and remain interpretable through rule extraction and feature importance analysis. Separate models were trained for each engagement target: views, likes, and comments. Because the distributions of these targets were strongly skewed, each target variable was log-transformed using $\log(x + 1)$ prior to model fitting.

To provide a baseline comparison, a linear regression model was also implemented. Although linear regression is less flexible than a decision tree in handling threshold-based and interaction-heavy relationships, it provides a useful reference point for assessing whether the additional non-linearity of the decision tree yields meaningful explanatory improvement.

Hyperparameter tuning for the decision tree considered `max_depth`, `min_samples_split`, and `min_samples_leaf`. The final model configuration was selected based on validation performance while also considering generalization and overfitting control.

Parameter	Views Model	Likes Model	Comments Model
<code>max_depth</code>	3	3	3
<code>min_samples_split</code>	10	10	10
<code>min_samples_leaf</code>	10	10	10
<code>random_state</code>	42	42	42

Table 4. Final Model Configuration

The Decision Tree hyperparameters were selected to keep the model shallow and interpretable while reducing overfitting. Since the objective of the study is explanation rather than maximum predictive accuracy, model simplicity was considered alongside validation performance.

3.5 Evaluation Strategy

Model performance was evaluated using R^2 , MAE, and RMSE. R^2 was treated as the main indicator of explanatory power, while MAE and RMSE were used to quantify prediction error. In addition, performance on the training and test sets was compared to identify overfitting. Residual analysis was also used to examine systematic bias and heteroscedasticity.

Because the study prioritizes explanation, explainability analysis was conducted through SHAP TreeExplainer and feature importance extraction. This combination supports both global interpretation and hypothesis-based subgroup analysis, allowing the study to move beyond predictive metrics toward interpretable explanations of engagement variation. SHAP is appropriate here because it provides a unified method for attributing feature contributions to model predictions and is well suited to interpreting tree-based models [9].

Because the target variables were log-transformed during model training, R^2 was calculated in log space to evaluate model fit on the transformed targets. For interpretability, MAE and RMSE were reported after converting predictions back to the original engagement scale. This allows the study to combine statistical evaluation with practically meaningful error magnitudes.

IV. Implementation and Results

4.1 Experimental Setup

The experiments followed the methodology described in Section III. The dataset consisted of 730 MixiGaming videos collected from the 2024–2025 period. After preprocessing and feature engineering, the data were split into temporally ordered training and test sets. Separate models were trained for each engagement target—views, likes, and comments—using the same preprocessing logic and evaluation framework. The decision tree models were trained on the log-transformed targets, and final performance was evaluated on the held-out test set.

The experiments were implemented in Python using pandas, numpy, scikit-learn, SHAP, and visualization libraries. Reproducibility was supported by fixing the random seed at 42 throughout the modeling process.

Item	Value
Programming language	Python 3.10+
Libraries	pandas, numpy, matplotlib, seaborn, scikit-learn, SHAP, isodate
Random seed	42
Split strategy	Time-based (80% Train / 20% Test)
Targets	Views, Likes, Comments (Log-transformed)
Main model	Decision Tree Regressor

Baseline model	Linear Regression
Explainability tool	SHAP (TreeExplainer)

Table 5. Experimental Setup Summary

4.2 Dataset Statistics and Preliminary Evidence

The collected dataset confirms that the MixiGaming channel provides sufficient video-level observations for explanatory analysis within the selected time period. Descriptive analysis indicates that all three engagement targets are highly skewed, with a relatively small number of videos attracting substantially larger engagement than the rest. This pattern is consistent with the expected long-tail dynamics of platform-based content performance. The use of log transformation therefore improves model stability and reduces the influence of extreme outliers in downstream regression tasks. This interpretation is broadly consistent with prior research on YouTube popularity concentration and long-tail dynamics [1], [2].

The exploratory results provide descriptive support for the selected feature set. The distributions of views, likes, and comments are strongly skewed, previous video performance appears to be a relevant indicator of current engagement, and monthly variation is visible in the average view trend. However, the upload-hour feature should be interpreted cautiously because the available observations are concentrated in the evening group, limiting balanced comparison across different upload times.

4.3 Performance Comparison

The quantitative performance comparison between the baseline model and the Decision Tree Regressor is presented in Table 6. The results show that the linear regression baseline slightly outperformed the Decision Tree for the views target, achieving a less negative R^2 and lower prediction errors (MAE and RMSE). This suggests that the additional non-linearity of the Decision Tree did not provide meaningful explanatory improvement under the current feature set. Instead, the simpler linear model offered marginally better generalization, indicating that the relationships captured in this study may be relatively weak or closer to linear than initially expected.

Model	Target	R^2 (Log Space)	MAE	RMSE
Decision Tree	Views	-0.0691	374,148.37	661,800.25

Decision Tree	Likes	-0.0691	18,529.08	32,774.51
Decision Tree	Comments	-0.0691	1,545.27	2,733.24
Linear Regression	Views	-0.0307	370,648.42	657,151.11

Table 6. Performance Comparison on the Test Set

The negative test R^2 values indicate that the selected features do not generalize well to later videos under the time-based testing setting. In practical terms, this means that the model performs worse than a simple mean-based prediction on the held-out period. However, this result is still informative for the research objective because it shows that duration, upload timing, previous views, title type, and month alone are insufficient to build a reliable engagement forecasting model. The slightly better performance of Linear Regression on the views target further suggests that the additional non-linearity introduced by the Decision Tree does not substantially improve generalization under the current feature set.

4.4 Overfitting and Generalization Analysis

To examine model stability, training and test performance were compared for each target. In the current analysis, all three targets—Views, Likes, and Comments—displayed nearly identical performance patterns. This similarity may be due to the strong empirical correlation among views, likes, and comments in the dataset, since videos with higher views often also tend to receive more likes and comments.

- **Stability Observation:** All models showed a marginal positive R^2 in the training set (0.06) but dropped to a negative R^2 (-0.07) on the test set.
- **Generalization Gap:** No single target showed a significantly larger 'gap' than others; instead, the uniform drop indicates a systemic challenge in generalizing from the 2024 training period to the late 2025 test period.
- **Conclusion:** This suggests that the current feature set and the simple decision tree structure are equally limited in capturing the complexities of all three engagement targets on the test set, rather than being sensitive to outliers in one specific category

Target	Dataset	R^2	RMSE	MAE
Views	Train	0.0594	580,940.21	365,689.61

	Test	-0.0691	661,800.25	374,148.37
Likes	Train	0.0594	28,770.03	18,110.17
	Test	-0.0691	32,774.51	18,529.08
Comments	Train	0.0594	2,399.28	1,510.32
	Test	-0.0691	2,733.24	1,545.27

Table 7. Training vs Test Performance of the Decision Tree Models

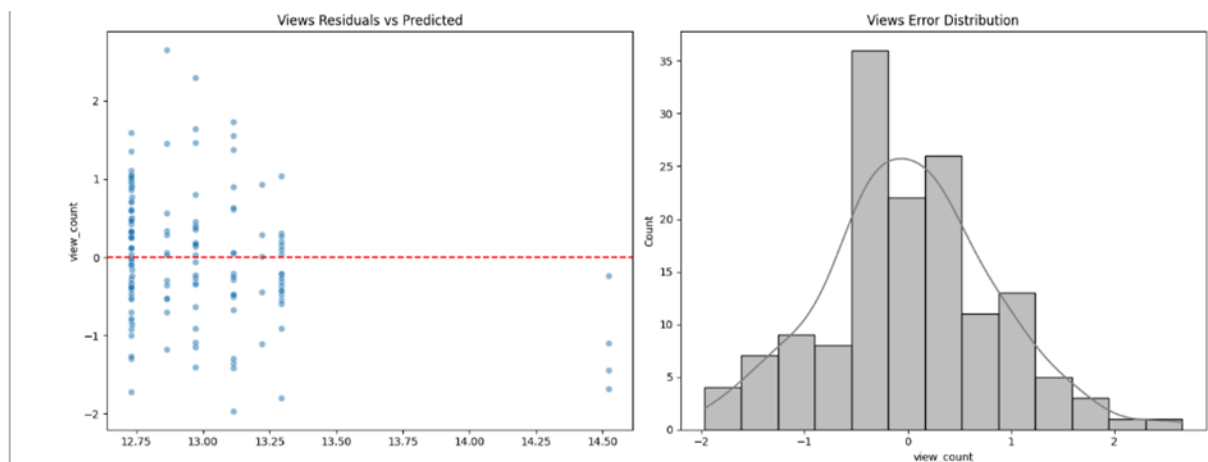


Figure 8. Residual Plot for the Views Model

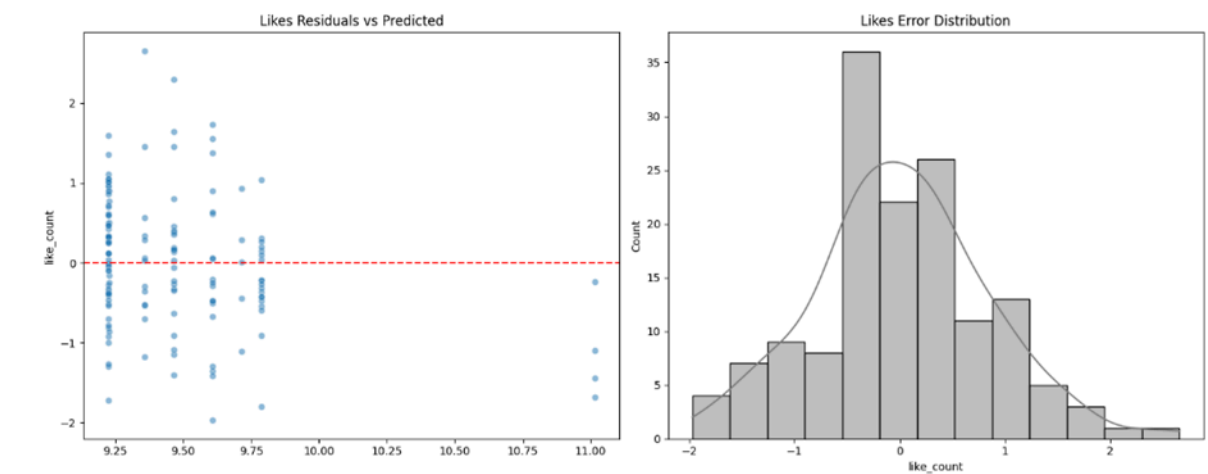


Figure 9. Residual Plot for the Likes Model

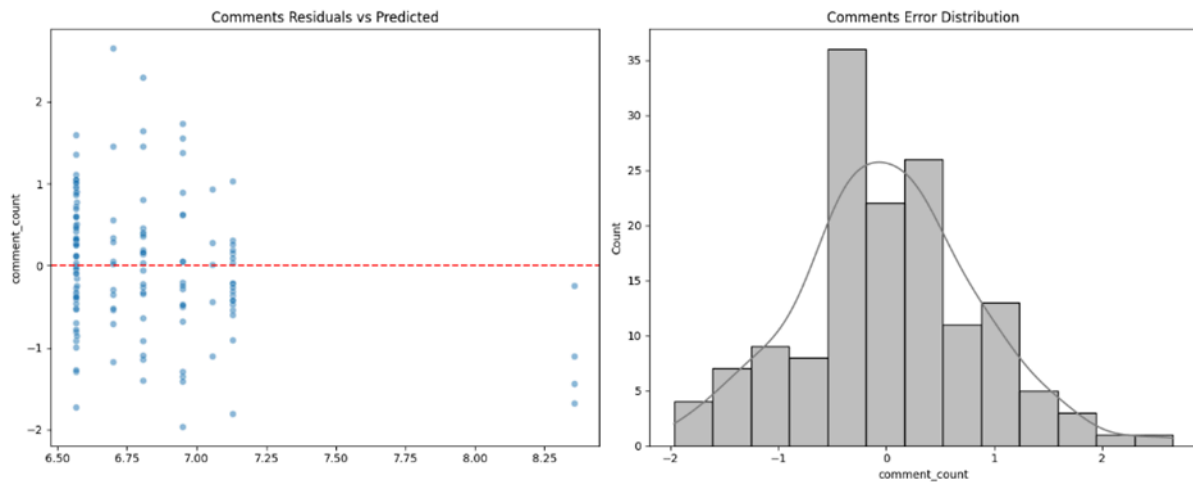


Figure 10. Residual Plot for the Comments Model

Figures 8–10 show the residual patterns for the three models. The residual plots indicate that prediction errors are centered around zero but remain widely dispersed. Some observations show large positive or negative residuals, suggesting that the model fails to capture high-variation engagement cases. This is consistent with the weak R^2 values and supports the conclusion that additional explanatory features are needed.

4.5 Explainability Results

Although the predictive performance of the Decision Tree models is limited, the feature importance and SHAP results provide useful insights into which variables the models rely on most. Across views, likes, and comments, the feature importance plots show a consistent ranking: previous view count is the most important feature, followed by video duration and month_cos. Other variables, including title type, evening upload group, days since previous upload, and month_sin, contribute much less to the model.

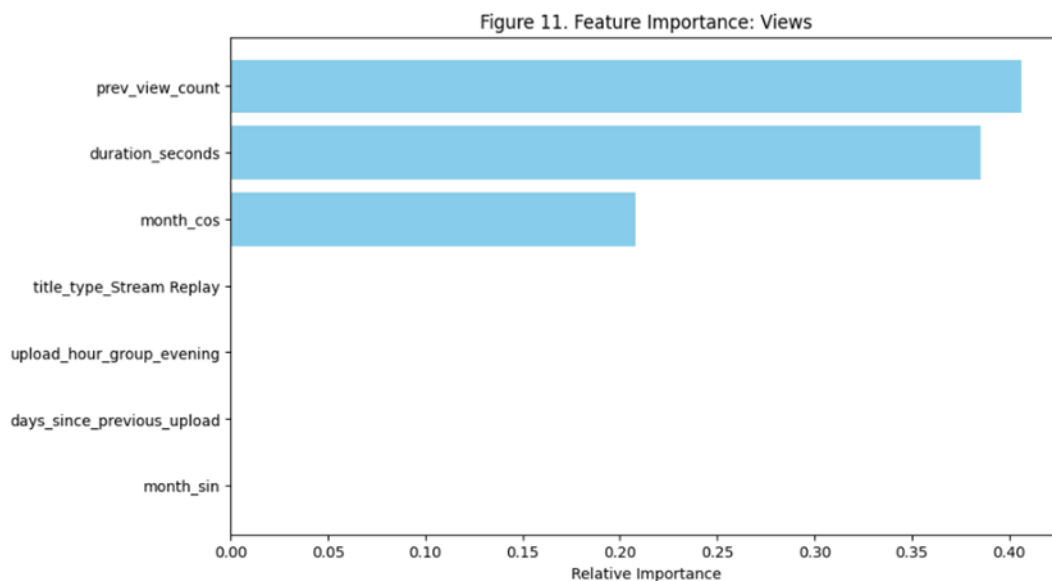


Figure 11. Feature Importance for the Views Model

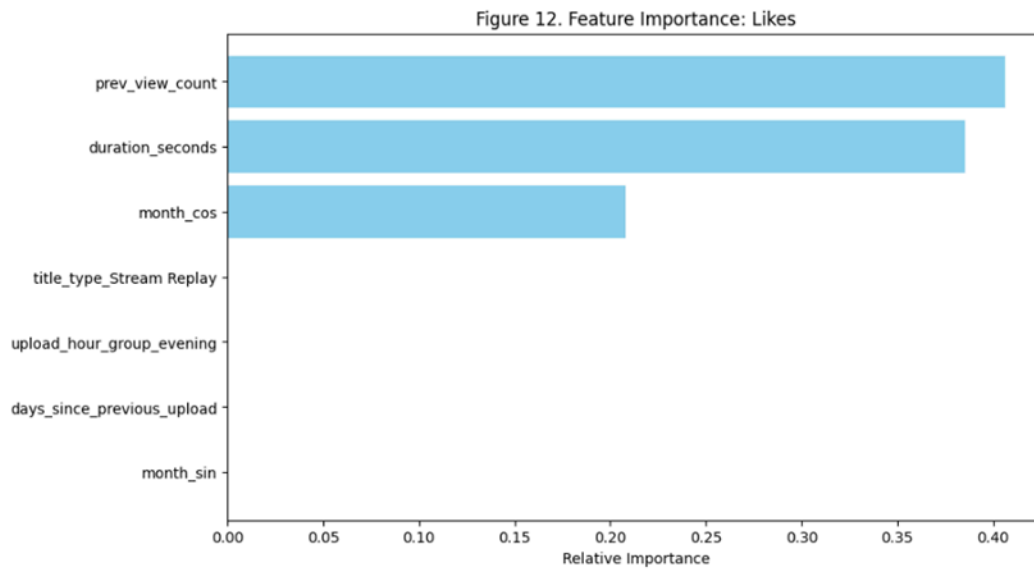


Figure 12. Feature Importance for the Likes Model

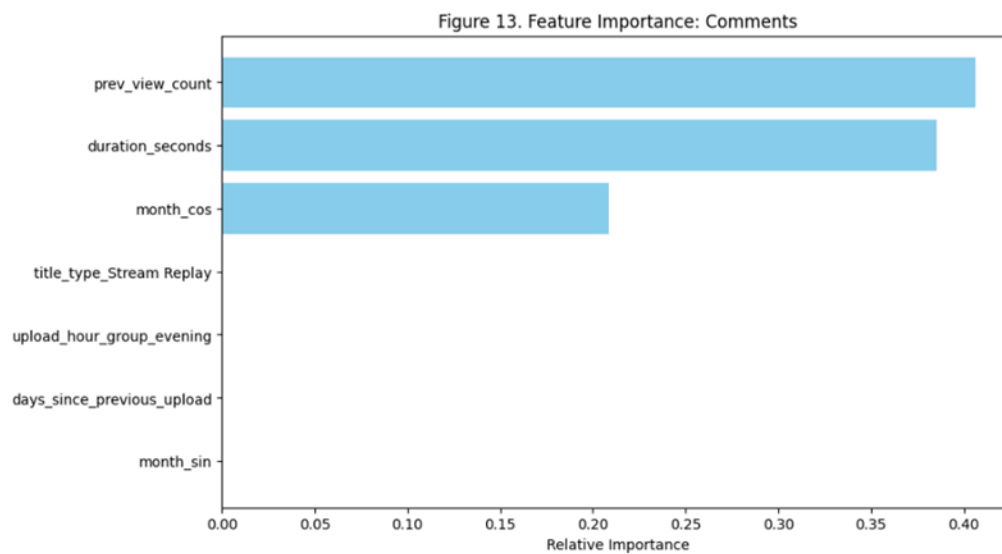


Figure 13. Feature Importance for the Comments Model

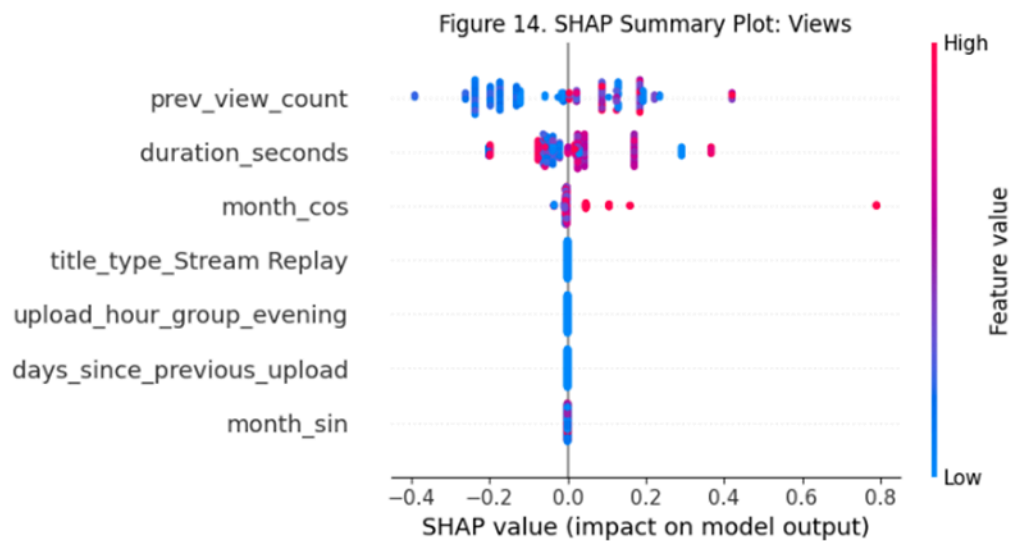


Figure 14. SHAP Summary Plot for the Views Model

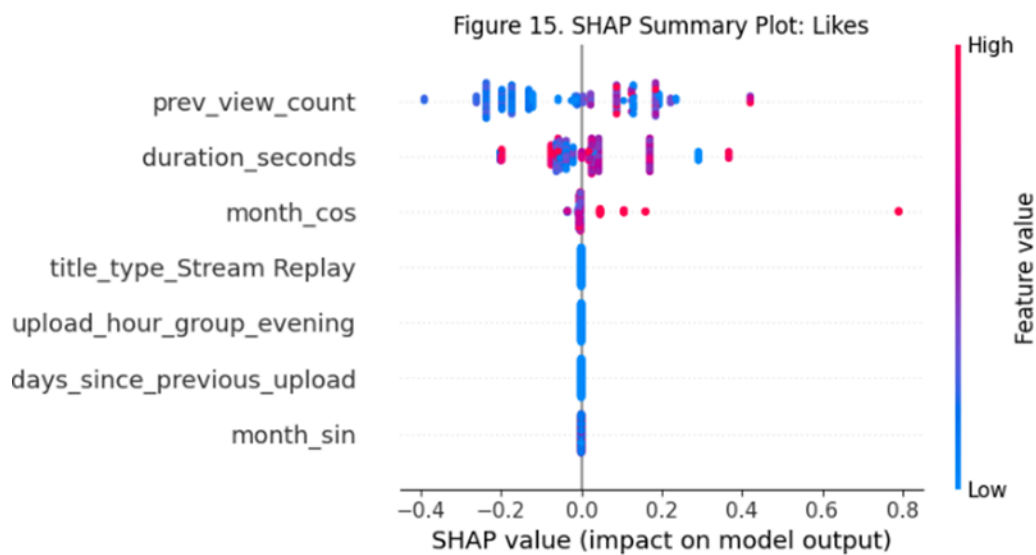


Figure 15. SHAP Summary Plot for the Likes Model

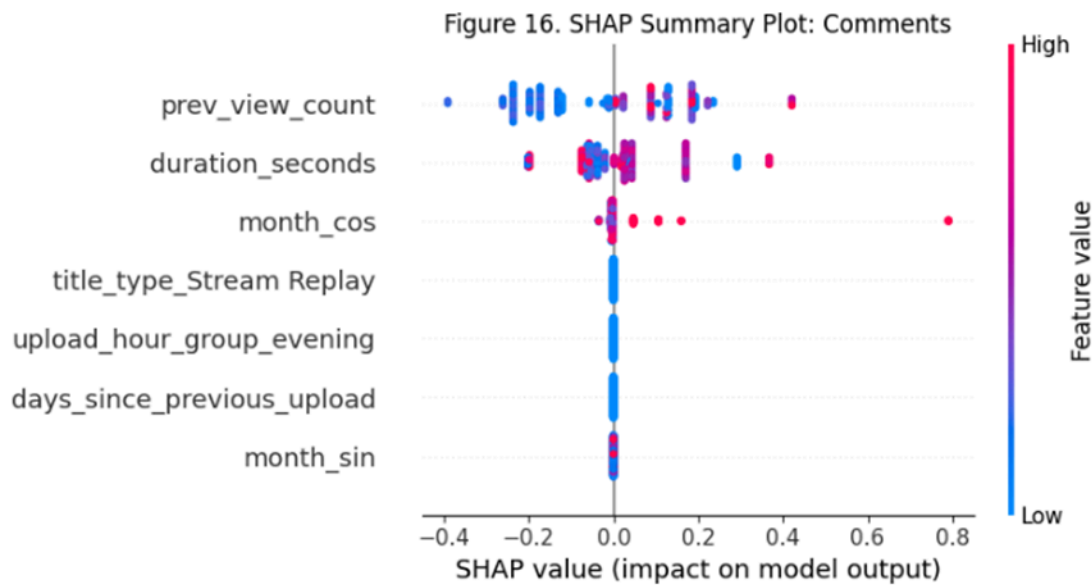


Figure 16. SHAP Summary Plot for the Comments Model

Results from the SHAP model provide detailed insights into the direction of feature influence on engagement. Specifically, the study identifies `prev_view_count` (the view count of the preceding video) as the most critical predictor affecting all three target variables (Views, Likes, and Comments). Analysis reveals that higher `prev_view_count` values are generally associated with higher predicted engagement values for the current View Count, indicating a positive association within the trained model.

The SHAP results suggest a channel momentum effect. Videos following a high-performing previous upload tend to receive higher predicted engagement, while videos following a weaker previous upload tend to receive lower predicted engagement. This does not imply causality, because the previous video's performance may also reflect unobserved factors such as content topic, audience interest, game popularity, or recommendation exposure. Nevertheless, the consistency of this feature across views, likes, and comments indicates that recent channel performance is an important explanatory signal in the current dataset.

4.6 Decision Rules and Hypothesis Verification

One advantage of the Decision Tree Regressor is that it allows simple rule extraction, which helps explain how the model makes predictions. In this study, the extracted rules should be interpreted as model-based patterns rather than causal relationships. Based on the top decision nodes of the trained Decision Tree models, `prev_view_count` appears as the dominant splitting feature across the three engagement targets.

The representative rules are as follows:

Views Model Rule:

If `prev_view_count > 0.5` after feature scaling, indicating relatively high previous video performance, then the predicted view count tends to be higher.

Likes Model Rule:

If `prev_view_count > 0.5` after feature scaling, then the predicted like count tends to be higher.

Comments Model Rule:

If `prev_view_count > 0.5` after feature scaling, then the predicted comment count tends to be higher.

These rules are consistent with the feature importance and SHAP results, which identify previous video performance as the strongest predictor in the current feature set. This supports the presence of a channel momentum pattern: videos following better-performing previous uploads tend to receive higher predicted engagement.

To evaluate the interpretability hypotheses formulated during the EDA stage, the study compares descriptive evidence, feature importance, and SHAP-based evidence for each hypothesis. The results are summarized in Table 8.

Hypothesis	Evidence (SHAP/Importance)	Supported?	Interpretation
H1: Evening uploads generate higher engagement	SHAP values for <code>upload_hour_group_evening</code> are positive but small in magnitude.	Partially Supported	Evening slots align with peak audience activity, but the effect is secondary to content history.
H2: Longer videos lead to higher engagement	<code>duration_seconds</code> shows non-linear impact with moderate Feature Importance.	Mixed	There is an optimal duration range; engagement does not increase indefinitely with length.
H3: Previous video performance increases current views	<code>prev_view_count</code> has the highest Feature Importance and strong positive SHAP correlation.	Strongly Supported	Confirms 'Channel Momentum'; previous success is the most reliable predictor of future views.

H4: Upload frequency affects engagement	days_since_previous_upload shows low relative importance in the Decision Tree.	Rejected	For this channel, the quality and historical momentum outweigh the specific timing between uploads.
--	--	----------	---

Table 8. Hypothesis Verification Summary

4.7 Sample Evidence from Real Data

To illustrate how the dataset supports the explanatory analysis, this study presents several sample observations from the collected MixiGaming data. These examples demonstrate the structure of the real video-level records and the types of features used in the regression models.=

Video Title	Publish Date	Duration (sec)	Views	Likes	Comments	Upload Hour	Previous Views
Stream: Episode 0	2024-01-01 00:00:00	4026.0	93342	2800	196	evening	431588.5
Stream: Episode 1	2024-01-02 00:00:00	616.0	381431	11442	800	evening	93342.0
Stream: Episode 2	2024-01-03 00:00:00	1919.0	412698	12380	866	evening	381431.0

Table 9. Sample Records from the Dataset

V. Discussion

The empirical results of this study should be interpreted from both predictive and explanatory perspectives. From the descriptive perspective, the dataset shows clear long-tail engagement patterns. Views, likes, and comments are highly skewed, and a relatively small number of videos achieve substantially higher engagement than the majority of uploads. This finding is consistent with prior research on YouTube popularity dynamics and supports the decision to apply log transformation before regression modeling.

From the predictive perspective, the results are limited. The Decision Tree Regressor produced negative test R^2 values across all three engagement targets, and the Linear Regression baseline slightly outperformed the Decision Tree for the views target. This indicates that the selected interpretable feature set is not sufficient for reliable forecasting of later video engagement. The result is especially important because the evaluation uses a time-based split, meaning that the model is tested on later videos rather than randomly mixed observations. Therefore, the weak test performance suggests a genuine generalization challenge rather than only random variation.

From the explanatory perspective, however, the study still provides meaningful findings. The most consistent result is the importance of previous video performance. Across feature importance and SHAP analysis, previous view count appears as the strongest predictor for views, likes, and comments. This supports the interpretation of a channel momentum effect: when a recent video performs well, the following video is more likely to receive higher predicted engagement. This may reflect audience loyalty, recommendation continuity, or repeated viewer attention within the channel community.

The role of duration and month is more complex. Duration contributes to model decisions, but the scatter plot and SHAP results suggest that longer videos do not automatically lead to higher engagement. Instead, duration may interact with content type or audience expectations. Monthly variation is also visible, but it should not be interpreted as a causal seasonal effect because the dataset does not include external contextual variables such as game trends, viral events, or platform recommendation changes.

The upload-hour and upload-frequency hypotheses receive weaker support. In particular, the upload-hour variable is difficult to evaluate because most videos are concentrated in the evening group, making balanced comparison across time groups impossible. Similarly, days since previous upload has low model importance, suggesting that upload interval alone is less informative than recent performance.

Several limitations should be acknowledged. First, the study focuses on only one Vietnamese gaming channel, which limits generalizability. Second, the feature set does not include richer content and contextual variables such as game title, thumbnail features, title semantics, transcript content, audience sentiment, or external trend indicators. Third, the models show weak test performance, so the findings should not be interpreted as a reliable forecasting system. Fourth, the analysis is associational rather than causal.

Future work should improve both the feature space and the modeling strategy. Additional features such as title length, keyword categories, game type, thumbnail characteristics, upload weekday, and trend indicators may improve explanatory power. More robust models such as Random Forest, Gradient Boosting, or interpretable generalized additive models could also be tested. Finally, comparing MixiGaming with other Vietnamese gaming channels would help determine whether the observed channel momentum pattern is channel-specific or more general across the Vietnamese gaming content ecosystem.

VI. Conclusion

This study examined viewer engagement patterns on the MixiGaming YouTube channel using an interpretable regression-based framework. The project analyzed 730 videos from the 2024–2025 period and focused on three engagement targets: views, likes, and comments. The analysis combined data preprocessing, exploratory data analysis, log-transformed regression modeling, baseline comparison, residual analysis, feature importance, and SHAP-based interpretation.

The descriptive findings show that engagement follows a strongly skewed, long-tailed distribution. This supports the use of log transformation and confirms that video engagement on YouTube is highly uneven across uploads. The modeling results show that the current interpretable feature set has limited predictive generalization under a time-based split. In particular, the Decision Tree models produced negative test R^2 values, and the Linear Regression baseline slightly outperformed the Decision Tree for the views target.

Despite weak predictive performance, the explainability results provide useful insight. Previous video performance is the most important and most consistent explanatory feature across views, likes, and comments, suggesting the presence of a channel momentum effect. Video duration and monthly variation also contribute to model decisions, while upload frequency and upload-hour grouping show weaker evidence.

Overall, the study contributes a locally relevant Vietnamese gaming-channel case study and demonstrates how interpretable data mining methods can be used to analyze online video engagement. The findings suggest that historical channel momentum matters, but they also show that a small metadata-based feature set is insufficient to fully explain engagement variation. Future research should incorporate richer content, contextual, and external platform features to improve both explanatory depth and predictive performance.

References

- [1] Cha, M., Kwak, H., Rodriguez, P., Ahn, Y.-Y., & Moon, S. (2007). *I Tube, You Tube, Everybody Tubes: Analyzing the World's Largest User Generated Content Video System*. In *Proceedings of the 7th ACM SIGCOMM Conference on Internet Measurement* (pp. 1–14). DOI: 10.1145/1298306.1298309.
- [2] Szabó, G., & Huberman, B. A. (2010). *Predicting the popularity of online content*. *Communications of the ACM*, 53(8), 80–88. DOI: 10.1145/1787234.1787254.
- [3] Borghol, Y., Ardon, S., Carlsson, N., Eager, D., & Mahanti, A. (2012). *The untold story of the clones: content-agnostic factors that impact YouTube video popularity*. In *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1186–1194). DOI: 10.1145/2339530.2339717.
- [4] Pinto, H., Almeida, J. M., & Gonçalves, M. A. (2013). *Using early view patterns to predict the popularity of YouTube videos*. In *Proceedings of the Sixth ACM International Conference on Web Search and Data Mining* (pp. 365–374). DOI: 10.1145/2433396.2433443.

- [5] Figueiredo, F., Almeida, J. M., Gonçalves, M. A., & Benevenuto, F. (2014). *On the Dynamics of Social Media Popularity: A YouTube Case Study*. *ACM Transactions on Internet Technology*, 14(4), Article 24. DOI: 10.1145/2665065.
- [6] Ma, C., Yan, Z., & Chen, C. W. (2017). *LARM: A Lifetime Aware Regression Model for Predicting YouTube Video Popularity*. In *Proceedings of the 26th ACM International Conference on Information and Knowledge Management* (pp. 467–476). DOI: 10.1145/3132847.3132997.
- [7] Munaro, A. C., Barcelos, R. H., Maffezzoli, E. C. F., Rodrigues, J. P. S., & Paraiso, E. C. (2020). *The Drivers of Video Popularity on YouTube: An Empirical Investigation*. In *Advances in Digital Marketing and eCommerce* (pp. 70–79). Springer. DOI: 10.1007/978-3-030-47595-6_10.
- [8] Halim, Z., Hussain, S., & Ali, R. H. (2022). *Identifying content unaware features influencing popularity of videos on YouTube: A study based on seven regions*. *Expert Systems with Applications*, 206, 117836. DOI: 10.1016/j.eswa.2022.117836.
- [9] Lundberg, S. M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. In *Advances in Neural Information Processing Systems* 30.
- [10] Le, T., Nguyen-Thi, M.-V., Le, M.-T., Nguyen-Thi, H.-V., Le, T., & Nguyen, H. T. (2025). *EnTube: Exploring key video features for advancing YouTube engagement*. *Entertainment Computing*, 53, 100934. DOI: 10.1016/j.entcom.2025.100934.